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Development of a Computational Platform to Generate Holograms Using 2D Images

A Report

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Abstract

Computational holography enables the generation of holograms using numerical algorithms and mathematical models of light propagation. Unlike conventional holography, it allows holographic patterns to be generated digitally. In this project, we have developed a computational platform to generate holograms using two-dimensional (2D) images and study their corresponding reconstructed outputs.

We explored a workflow for learning-based phase prediction and optimization using PyTorch. The generated phase maps were examined, and the corresponding reconstructed images were compared with the original inputs to evaluate how effectively the visual information was preserved. We also studied the role of the Adam optimizer in the model training process.

The experimental observations highlighted that reconstruction quality depends on factors such as training data, phase representation, model optimization, and light propagation assumptions. Differences between the original and reconstructed images helped us identify the limitations of the current implementation and understand the challenges involved in achieving accurate reconstruction. The results also indicate scope for improving phase prediction, model performance, and overall reconstruction quality.

This project establishes a computational workflow for image-to-hologram generation and provides practical insight into learning-based computational holography. Future work may focus on improving the dataset, prediction model, phase optimization, and light propagation methods to enhance reconstruction quality.

Introduction

Background

Holography is a technique used to record and reconstruct light from an object. An ordinary photograph mainly records light intensity, whereas holography also considers the phase of light waves. The phase describes the relative position of a light wave within its cycle and plays an important role in reproducing the appearance of an object during reconstruction.

Computational holography uses computers and numerical algorithms to generate holograms. In Computer-Generated Holography (CGH), a computer calculates a suitable phase pattern that can produce a desired image when light propagates from the hologram plane. This allows holographic patterns to be investigated digitally before using physical optical equipment.

Problem Statement

Generating a suitable phase pattern from a given 2D image is a challenging task. The reconstructed image may differ from the original because of phase errors, speckle noise, imperfect optimization, and limitations in the numerical propagation model. Therefore, a computational workflow is needed to generate phase maps, reconstruct images, and study the quality of the resulting outputs.

Objectives

The main objectives of this project are:

- To develop a computational workflow for generating holographic phase patterns from 2D images.
- To understand phase generation, numerical light propagation, and image reconstruction.
- To explore phase optimization and CNN-based phase prediction using PyTorch.
- To compare reconstructed images with their original inputs and identify limitations in the implemented methods.

Scope

The project focuses on numerical hologram generation and reconstruction. It investigates computational phase generation, learning-based prediction, and reconstruction quality. Physical hologram display using a Spatial Light Modulator (SLM) and experimental optical reconstruction are outside the scope of the current implementation.

Methodology

The methodology consists of image preprocessing, phase-map generation, numerical propagation, reconstruction, and evaluation. The main objective is to study how a digitally generated holographic phase pattern can reproduce the target image.

1. Input Image Preparation

A 2D image is selected as the target and resized to a suitable computational resolution. The image is converted into a numerical array, and its intensity values are normalized before phase generation or model training.

2. Phase Generation

A holographic field can be represented as

$$U(x,y) = A(x,y)e^{i\phi(x,y)}$$

where $A(x,y)$ represents amplitude and $\phi(x,y)$ represents phase. The aim is to calculate a phase pattern that produces a reconstruction similar to the target image.

Three computational approaches are investigated:

- **Iterative Phase Retrieval:** An iterative Fourier-transform-based method updates the phase using constraints in the image and hologram planes. The process aims to improve the reconstructed intensity over successive iterations.
- **Direct Phase Optimization:** The phase map is treated as a trainable parameter. The reconstructed intensity is compared with the target using Mean Squared Error (MSE). The Adam optimizer updates the phase to reduce the reconstruction loss.
- **CNN-Based Phase Prediction:** A Convolutional Neural Network learns a mapping from input images to phase maps. The model uses training examples and a loss function to learn phase predictions.

3. Numerical Reconstruction

The generated phase pattern is used to form a complex field. Numerical propagation estimates the field at the reconstruction plane. The reconstructed intensity is calculated as

$$I(x,y) = |U(x,y)|^2$$

The reconstructed image is then compared with the original target to examine the preservation of its structure, brightness, and details.

Results and Discussion

The experiments demonstrate that computational methods can generate phase patterns and produce numerical reconstructions from 2D images. However, reconstruction quality varies according to the phase-generation method and the optimization process.

1. Iterative Phase Retrieval

The iterative method generated a phase pattern with rapidly varying phase values. The reconstructed image preserved some of the main object structures, but fine details and intensity values differed from the original. This indicates that iterative phase retrieval can produce useful reconstructions, although errors may remain.

2. Direct Phase Optimization

In direct phase optimization, the phase map was updated using the Adam optimizer to reduce the difference between the reconstructed and target images. In the recorded experiment, the loss decreased from approximately 0.754 to 0.0333 at iteration 1200. This reduction indicates improvement according to the selected loss function. However, speckle noise and visual differences remained in the reconstructed output.

3. CNN-Based Phase Prediction

The CNN experiment used 2000 images, with 1600 images for training and 400 for validation. The model was trained for 25 epochs using a batch size of 8.

Table 1: CNN training and validation losses

Epoch	Training Loss	Validation Loss
1	0.026832	0.025882
5	0.024877	0.024907
10	0.025154	0.025974
15	0.022956	0.024573
20	0.022567	0.023186
25	0.022161	0.022906

The final training loss was 0.022161, while the final validation loss was 0.022906. The lowest recorded training loss was 0.021522 at epoch 24, and the lowest validation loss was 0.022278 at epoch 23. The reconstructed output preserved the coarse image shape, but blurring and ringing effects remained.

Discussion

The experiments show that a lower loss does not necessarily guarantee a visually accurate reconstruction. Image quality depends on the input data, phase representation, optimization, and propagation assumptions.

Conclusion

This project investigated computational hologram generation from 2D images through phase generation, numerical propagation, and image reconstruction. Learning-based phase prediction and direct phase optimization were explored using PyTorch. The generated phase maps and reconstructed images provided practical insight into the computational process and its limitations.

The experiments showed that different approaches preserve different aspects of the target image. Iterative phase retrieval retained some major image structures, direct phase optimization reduced the selected reconstruction loss, and the CNN learned to predict phase patterns from input images. However, the reconstructed outputs still showed limitations such as blur, speckle noise, ringing, and differences in image intensity and fine details.

The results indicate that reconstruction quality depends on the input data, phase-generation method, optimization process, and assumptions used for numerical light propagation. The experiments also demonstrate that numerical loss values alone are insufficient to establish the visual quality of a reconstructed hologram.

Limitations

The current implementation is limited by the available training data, model design, phase representation, and numerical propagation assumptions. The project focuses on computational reconstruction and does not demonstrate physical optical reconstruction using an SLM or camera.

Future Scope

Future work may include expanding the dataset, improving the CNN architecture, experimenting with alternative loss functions, and refining numerical propagation methods. The reconstructed outputs may be evaluated using Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). Finally, integration with a Spatial Light Modulator and camera-based feedback could help evaluate generated holograms in a physical optical setup.